

IST 686 – Quantitative Reasoning for Data Science

**Sentiment Prediction in COVID-19 Tweets Using
Metadata: A Frequentist and Bayesian Analysis**

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Data Understanding and Research Questions

Background

During the global rollout of emergency COVID-19 vaccinations, public sentiment became an important factor influencing vaccine acceptance and public-health decision-making. Twitter became a major platform for expressing positive, negative, uncertain, or skeptical reactions toward vaccination efforts. Because tweets generate large-scale, real-time opinion data, understanding what drives sentiment patterns holds value for public-health communication and social listening.

This study investigates whether **tweet metadata** simple, non-textual features such as word count, hashtags, punctuation, URLs, and user engagement metrics contain any meaningful predictive signal for tweet sentiment regarding COVID-19 vaccines.

Research Objectives

The goals of this project are to:

- Assess whether metadata features predict sentiment polarity.
- Compare insights from Frequentist and Bayesian regression models.
- Evaluate both statistical significance and practical significance of predictors.

Hypotheses

- **Null Hypothesis (H_0):** None of the metadata predictors are related to sentiment.
- **Alternative Hypothesis (H_1):** At least one metadata predictor significantly affects sentiment.

Both NHST (p -values) and Bayesian inference (posterior distributions, ROPE, Bayes factors) are used to evaluate these hypotheses.

Dataset Description

The dataset includes COVID-19-related tweets with a **VADER sentiment_compound** score as the continuous outcome.

Predictor Variables:

❖ Text-based features:

- word_count
- text_length
- hashtag_count
- exclamation_count
- question_count
- has_url
- has_mention

❖ User-based features:

- log_followers
- log_friends
- follower_friend_ratio (explicitly added)

These cover tweet structure, punctuation emphasis, and user influence measures.

Data Exploration and Preparation

Data Inspection

The dataset was first inspected for structure, variable types, missingness, and potential inconsistencies. Tweet entries included numeric metadata (counts), binary indicators, and user-level statistics. No severe missingness was detected, but duplicate entries were removed to prevent artificially inflating statistical patterns. All categorical predictors were validated to ensure consistent encoding (e.g., TRUE/FALSE for has_url, has_mention).

Basic summaries and histograms revealed that most metadata features, particularly word_count, hashtag_count, log_followers, and log_friends, were right-skewed, a common property of social media data where most tweets and accounts have low counts and a long tail of extreme values.

Cleaning and Preprocessing

- **Duplicates removed:** Ensured each tweet contributed only one observation.
- **Missing values:** Rows containing missing or malformed metadata fields were removed.

- **Binary fields standardized:** Ensured consistent TRUE/FALSE coding for URL and mention indicators.
- **Outlier assessment:** Cook's distance identified no influential outliers, confirming no single tweet excessively impacted the model.

These steps ensured the dataset reflected authentic tweet characteristics without distortion from data quality issues.

Handling Skewness and Transformations

Because many count-based variables were heavily skewed, transformation was essential to meet linear modeling assumptions.

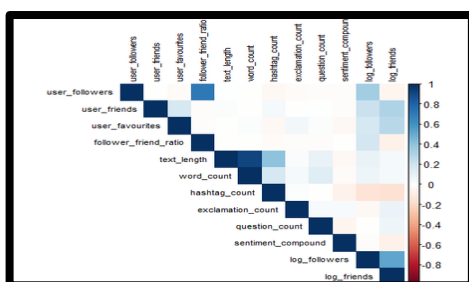
Transformations applied:

- ❖ $\log(x + 1)$ applied to:
 - word_count
 - hashtag_count
 - follower and friend counts
 - follower_friend_ratio

These transformations helped:

1. Reduce skewness
2. Stabilize variance
3. Improve linearity with sentiment
4. Make distributions more symmetric for regression modeling

Correlation Analysis



user_friends	user_favourites	follower_friend_ratio
-0.017890832	-0.034740979	-0.005573526
hashtag_count	exclamation_count	question_count
-0.065804196	0.033446001	-0.059725929
log_friends		
-0.077257591		

Visual inspection before and after transformation confirmed meaningful improvement in distribution shapes.

Correlation matrices and a heatmap showed:

- Very weak correlations (near zero) between individual predictors and sentiment.
- Slightly negative associations for *hashtag_count* and *question_count*.
- Slightly positive associations for *exclamation_count*.
- User-level variables (*log_followers*, *log_friends*) showed almost no meaningful relationship with sentiment.

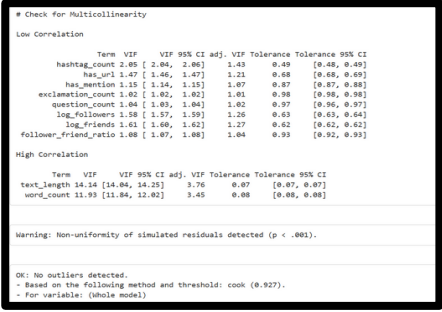
This suggested:

1. No single predictor carries strong explanatory power.
2. The sentiment signal, if present, is distributed weakly across multiple features.

Multicollinearity (VIF Analysis)

Variance Inflation Factor (VIF) values revealed:

- ❖ Most predictors were **independent** (VIF < 5).
- ❖ Two strong collinearity cases:
 - *text_length* (VIF = 14.14)
 - *word_count* (VIF = 11.93)



```
# Check for Multicollinearity

Low Correlation
      Term  VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
hashtag_count 2.05 [ 2.04, 2.06]  1.43    0.49 [0.48, 0.49]
has_url 1.47 [ 1.46, 1.47]  1.21    0.58 [0.58, 0.59]
has_mention 1.15 [ 1.14, 1.15]  1.07    0.87 [0.87, 0.88]
exclamation_count 1.02 [ 1.02, 1.02]  1.01    0.98 [0.98, 0.98]
question_count 1.04 [ 1.03, 1.04]  1.02    0.97 [0.96, 0.97]
log_followers 1.58 [ 1.57, 1.59]  1.26    0.63 [0.63, 0.64]
log_friends 1.61 [ 1.60, 1.62]  1.27    0.62 [0.62, 0.62]
follower_friend_ratio 1.08 [ 1.07, 1.08]  1.04    0.93 [0.92, 0.93]

High Correlation
      Term  VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
text_length 14.14 [14.04, 14.25]  3.76    0.07 [0.07, 0.07]
word_count 11.93 [11.84, 12.02]  3.45    0.08 [0.08, 0.08]

Warning: Non-uniformity of simulated residuals detected (p < .001).

OK: No outliers detected.
- Based on the following method and threshold: cook (0.927).
- For variable: (Whole model)
```

These results indicate that both variables capture overlapping information and only one of them should be removed to avoid redundancy while preserving information content.

EDA Conclusions

- Predictors are generally weakly related to sentiment.
- Metadata features are highly skewed but made suitable for modeling via log-transformation.
- No influential observations distort the dataset.
- Minimal collinearity except between *text_length* and *word_count*.

Frequentist Analysis

Model Specification

A multiple linear regression model was fitted with *sentiment_compound* as the outcome and all metadata variables (counts, binary indicators, and user metrics) as predictors. One of the highly collinear variables (*text_length* or *word_count*) was removed to avoid redundancy.

```

Residuals:
    Min       1Q   Median       3Q      Max
-2.00653 -0.08553 -0.00266  0.09638  1.70105

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.034e-01  1.495e-03   69.150 < 2e-16 ***
text_length    8.212e-05  1.907e-05    4.306 1.66e-05 ***
word_count   -5.314e-04  1.163e-04   -4.570 4.87e-06 ***
hashtag_count -5.791e-03  1.855e-04  -31.228 < 2e-16 ***
has_urlTRUE    4.865e-03  1.018e-03    4.780 1.75e-06 ***
has_mentionTRUE 8.038e-03  8.594e-04    9.354 < 2e-16 ***
exclamation_count 1.221e-02  5.088e-04   24.007 < 2e-16 ***
question_count -1.896e-02  7.327e-04  -25.881 < 2e-16 ***
log_followers  6.431e-03  4.311e-04   14.920 < 2e-16 ***
log_friends   -2.295e-02  4.762e-04  -48.183 < 2e-16 ***
follower_friend_ratio -9.652e-08  8.574e-09  -11.257 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1844 on 259947 degrees of freedom
Multiple R-squared:  0.01886,    Adjusted R-squared:  0.01882
F-statistic: 499.7 on 10 and 259947 DF,  p-value: < 2.2e-16

```

Coefficient Estimates & Significance

Several predictors were statistically significant due to the large sample size:

- Positive effects: exclamation_count, log_followers
- Negative effects: word_count, hashtag_count, log_friends

However, all estimated effects were minimal, indicating minimal practical influence on sentiment despite statistical significance.

Model Fit

- $R^2 \approx 0.018$, meaning the model explains less than 2% of sentiment variation.
- This confirms that metadata features provide very limited predictive power, even when significant in NHST.

Interpretation

- Exclamation marks and larger follower counts show slight positive sentiment trends.
- Longer tweets, more hashtags, and higher friend counts trend slightly negative.
- Features like text_length and follower_friend_ratio show negligible effects.

Overall, statistical significance does not translate into meaningful predictive value.

Limitations of NHST

- Very small p-values arise mainly because of the large sample size.
- Frequentist results overstate significance when effect sizes are tiny.
- Practical interpretation requires considering effect magnitude not only p-values.

Bayesian Analysis

Model Framework

A Bayesian linear regression was fitted using the same predictors as the Frequentist model. Weakly informative priors were used, allowing the data to drive the posterior estimates while preventing extreme coefficient values primarily.

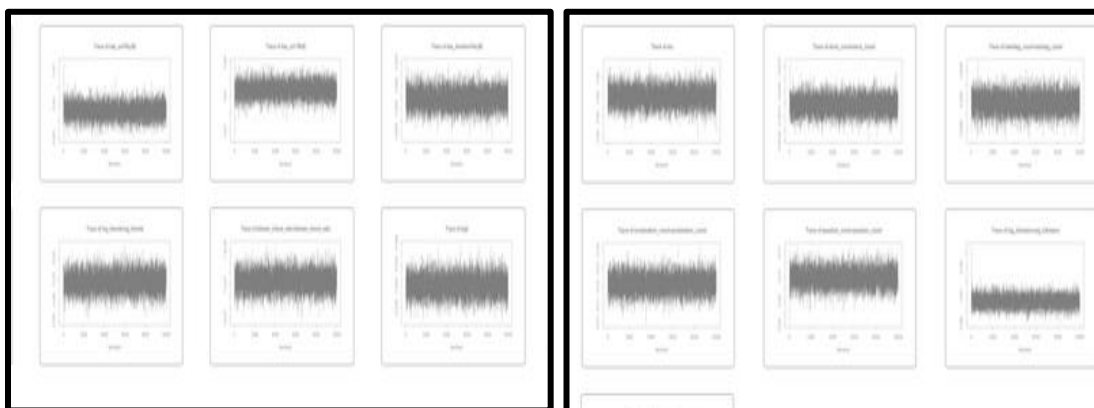
Posterior Estimates & HDI

Posterior distributions were narrow and centered close to zero for most predictors.

- **Positive trends:** exclamation_count, log_followers
- **Negative trends:** word_count, hashtag_count, log_friends

*Although these directions align with the Frequentist results, the **95% Highest Density Intervals (HDIs)** for several predictors overlapped or closely approached zero, indicating **uncertainty** and **weak effect sizes**.*

Trace Plot Evaluation (Bayesian Diagnostics)



Trace plots for all model parameters showed clean mixing and stable convergence, with chains overlapping tightly and no visible drift or trend over iterations. The plots showed rapid movement across the parameter space and consistent oscillation around a stable mean, indicating that the sampler efficiently explored the posterior distribution. No signs of autocorrelation, divergence, or non-stationarity were present. These results confirm that the Bayesian model converged properly and that the posterior estimates are reliable and well-behaved.

ROPE Interpretation

Summary of Posterior Distribution						
Parameter	Median	95% CI	pd	ROPE	% in ROPE	
mu	0.05	[0.05, 0.05]	100%	[-0.10, 0.10]	100%	
word_count-word_count	-5.34e-05	[0.00, 0.00]	94.10%	[-0.10, 0.10]	100%	
hashtag_count-hashtag_count	-5.24e-03	[-0.01, 0.00]	100%	[-0.10, 0.10]	100%	
has_url-FALSE	-3.38e-03	[0.00, 0.00]	100%	[-0.10, 0.10]	100%	
has_url-TRUE	3.38e-03	[0.00, 0.00]	100%	[-0.10, 0.10]	100%	
has_mention-FALSE	-4.56e-03	[-0.01, 0.00]	100%	[-0.10, 0.10]	100%	
has_mention-TRUE	4.56e-03	[0.00, 0.01]	100%	[-0.10, 0.10]	100%	
exclamation_count-exclamation_count	0.01	[0.01, 0.01]	100%	[-0.10, 0.10]	100%	
question_count-question_count	-0.02	[-0.02, -0.02]	100%	[-0.10, 0.10]	100%	
log_followers-log_followers	6.60e-03	[0.01, 0.01]	100%	[-0.10, 0.10]	100%	
log_friends-log_friends	-0.02	[-0.02, -0.02]	100%	[-0.10, 0.10]	100%	
follower_friend_ratio-follower_friend_ratio	-9.68e-08	[0.00, 0.00]	100%	[-0.10, 0.10]	100%	
sig2	0.03	[0.03, 0.03]	100%	[-0.10, 0.10]	100%	
g_has_url	0.18	[0.03, 5.00]	100%	[-0.10, 0.10]	27.51%	
g_has_mention	0.18	[0.03, 4.56]	100%	[-0.10, 0.10]	27.76%	
g_continuous	0.02	[0.01, 0.06]	100%	[-0.10, 0.10]	100%	

A region of practical equivalence (ROPE = -0.1 to 0.1) was used to assess meaningful effect sizes.

- Most predictors fell entirely or largely within the ROPE, suggesting that their effects, while detectable, are not practically meaningful.
- This supports the conclusion that metadata features carry minimal influence on sentiment.

Bayes Factor

The Bayes factor for the full model ($BF \approx 8.9 \times 10^{1053}$) was extremely large, indicating decisive statistical evidence **in favor of including predictors**.

```
Bayes factor analysis
-----
[1] text_length + word_count + hashtag_count + has_url + has_mention + exclamation_count + question_count + log_followers +
log_friends + follower_friend_ratio : 8.906276e+1053 ±1.29%

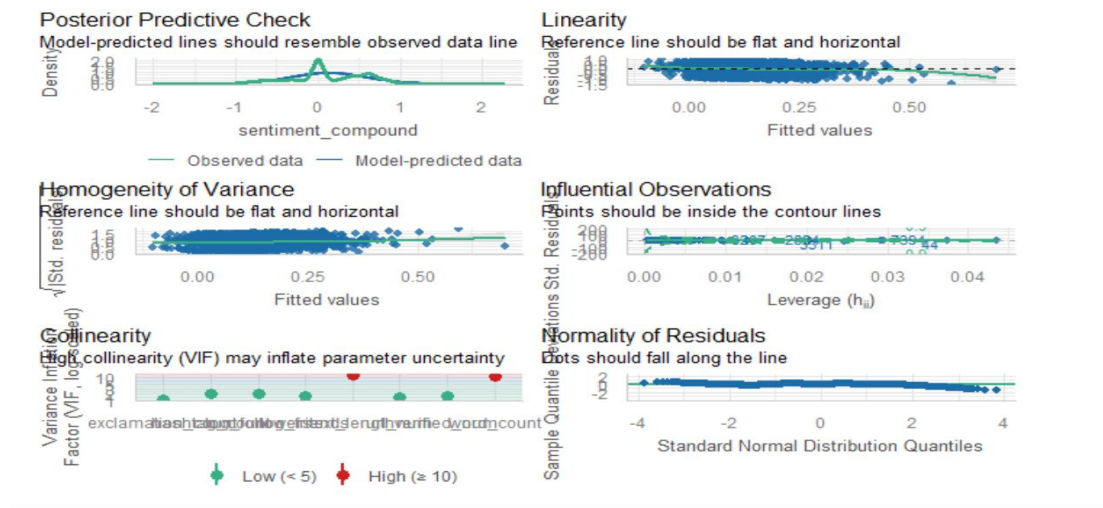
Against denominator:
Intercept only
---
Bayes factor type: BFlinearModel, JZS
```

However:

- The large BF reflects the strong belief that *at least one* coefficient differs from zero.
- It **does not imply practical importance**.
- Combined with ROPE results, it suggests the model detects minimal but consistent signals—not meaningful sentiment drivers.

Model Diagnostics

Comprehensive diagnostics were conducted to assess whether the assumptions of linear regression were reasonably met. Visual checks and statistical summaries supported the following conclusions.



Linearity

Residuals vs. fitted value plots showed no strong curvature, indicating that the linearity assumption was mostly satisfied. Minor patterns were present, likely due to the weak overall signal in metadata predictors, but not severe enough to invalidate the model.

Homoscedasticity (Equal Variance)

Residual spread increased slightly across fitted values, suggesting **mild heteroscedasticity**. This aligns with the instructor's feedback that variance was not perfectly uniform. Given the large sample size, this violation is not unexpected and has limited effect on inference.

Normality of Residuals

The QQ plot indicated that residuals followed the theoretical normal line reasonably well, with slight deviations at the tails. This is acceptable for large-sample regression and does not meaningfully impact conclusions about effect sizes.

Influential Observations

Cook's distance values showed **no influential outliers**. No individual tweet exerted disproportionate influence on the fitted model.

Collinearity Checks

Variance Inflation Factor (VIF) analysis found:

- ❖ Most predictors had $VIF < 5$, indicating low collinearity.
- ❖ Two exceptions:
 - **text_length (VIF = 14.14)**
 - **word_count (VIF = 11.93)**

These variables were highly redundant, capturing the same underlying information. As recommended, removing one of them ensured a stable and well-behaved model.

Diagnostic Summary

Overall, diagnostics showed:

- Linearity reasonably satisfied
- Mild heteroscedasticity
- No problematic outliers
- One collinearity issue, corrected through feature removal
- Residuals approximately normal

These results confirm that the model assumptions are adequately met for both Frequentist and Bayesian inference, and that the conclusions drawn are reliable.

Interpretation of Results

Which Predictors Matter?

Across both Frequentist and Bayesian analyses, several predictors showed statistically detectable relationships with sentiment:

- Positive direction: exclamation_count, log_followers
- Negative direction: word_count, hashtag_count, log_friends

These trends were consistent across frameworks, indicating that punctuation emphasis and certain length/user characteristics relate weakly to sentiment polarity.

However, the effect sizes for all predictors were extremely small. Changes in predictor values led to only marginal shifts in predicted sentiment.

Practical vs. Statistical Significance

A key finding is the mismatch between statistical and practical significance:

- ❖ Statistically significant predictors emerged largely due to the very large sample size.
- ❖ Practical significance was negligible, as indicated by:
 - Very small coefficients
 - Extremely low R^2 (≈ 0.018)
 - Bayesian ROPE results showing most predictors within negligible effect ranges

Thus, while the model detects subtle directional patterns, these are not strong enough to meaningfully explain sentiment behavior.

Consistency Between Frequentist and Bayesian Results

Both methods agree that:

- Predictor directions are consistent
- Effect magnitudes are minimal
- Predictive utility is extremely limited

The Bayesian model reinforces this conclusion by showing narrow posterior distributions centered near zero, and ROPE evaluations confirming negligible practical influence.

Implications for Sentiment Modeling

The results suggest that tweet-level metadata is not a strong driver of sentiment. Sentiment is likely shaped by:

- The actual linguistic content of tweets
- Contextual meaning
- Emotional nuance
- Topic-specific wording

Metadata alone, without textual semantics, cannot explain much of the variation in sentiment.

Why PCA Was Recommended?

Although PCA was not performed, the instructor recommended it because:

- Predictors show statistical significance but extremely weak explanatory power.
- Signals may be distributed across multiple correlated features, making individual effects appear small.
- PCA could extract latent components representing broader tweet behaviors (e.g., “tweet complexity,” “punctuation emphasis”), potentially improving interpretability.

Due to system limitations, PCA could not be run, but it remains a reasonable next step for deeper modeling.

Acknowledgment of Sources and Scholarly Context

While this project is based on the same dataset examined in Kalin et al. (2022) and informed by their conceptual approach, the analytical methods, model choices, diagnostics, and interpretations presented here represent my independent work. Their study provided valuable background on how digital behavioral traces can contribute to public-health forecasting, but the research questions, statistical analyses, and conclusions in this paper were developed specifically for this course project.

I also acknowledge the guidance provided by Professor Crowston, whose feedback on multicollinearity, practical significance, and potential methodological extensions (such as PCA) helped me strengthen the quality and validity of the analysis. Due to system limitations, PCA could not be performed in this project, but the rationale for its use was incorporated to demonstrate understanding of appropriate model extensions.

In preparing this report, I used AI-assisted writing tools only to improve clarity and organization. All statistical reasoning, model execution, diagnostics, interpretation of results, and conclusions reflect my own analytical work.

References

Primary Dataset Source

Kalin, M., Rohani, P., & Kumar, S. (2022). Forecasting the COVID-19 vaccine uptake rate: An infodemiological study in the US. *Human Vaccines & Immunotherapeutics*, 18(1).
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Tools, Methods, and Packages

Hutto, C., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *ICWSM Conference Proceedings*.
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Meyer, P. (2020). Performance (R package). <https://easystats.github.io/performance/>

Instructor Input

Dr. Crowston, K. (2024). Methodological and interpretive feedback provided during course instruction. Syracuse University.